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APPENDIX A

Computations in GAP

The calculations were performed in GAP ([4]).

A.1 16-cell

A.1.1 Cycles of the 16-cell

Distinguished generators of the 16-cell:

```
gap> r0:=(1,2)(5,6);
(1,2)(5,6)
gap> r1:=(2,3)(6,7);
(2,3)(6,7)
gap> r2:=(3,4)(7,8);
(3,4)(7,8)
gap> r3:=(4,8);
(4,8)
gap> P:=Group(r0,r1,r2,r3);
Group([ (1,2)(5,6), (2,3)(6,7), (3,4)(7,8), (4,8) ])
gap> eGC:=Elements(P);;
```

We will remove the involutions and the symmetries of order three (because these correspond to the triangular faces of the 16-cell and these cycles are in a plane).

```
gap> filP:=[];; ##Filter of P.
gap> for j in [1..Length(eGC)] do
>       if Order(eGC[j])=2 or Order(eGC[j])=3 then
>               continue;
>       else
>               Add(filP,eGC[j]);
>       fi;
> od;
gap> filP;;
gap> Length(filP);
277
gap> #Since the base edge is {1, 2} and the base vertex is 1, then we want a rotation
# that sends 1 to 2 when it rotates the first time.
```

```
gap> pHel:=[];;
gap> for i in [1..Length(filP)] do
>       if Length(Cycle(filP[i],1))>1 and Cycle(filP[i],1)[2]=2 then
>               Add(pHel,filP[i]);
>       fi;
> od;
gap> pHel;
[ (1,2)(3,4,7,8)(5,6), (1,2)(3,8,7,4)(5,6), (1,2,3)(4,8)(5,6,7),
  (1,2,3,4)(5,6,7,8), (1,2,3,4,5,6,7,8), (1,2,3,5,6,7),
  (1,2,3,5,6,7)(4,8), (1,2,3,8,5,6,7,4), (1,2,3,8)(4,5,6,7),
  (1,2,4,3)(5,6,8,7), (1,2,4,7,5,6,8,3), (1,2,4,5,6,8),
  (1,2,4,3,5,6,8,7), (1,2,4,7)(3,5,6,8), (1,2,4)(3,7)(5,6,8),
  (1,2,4,5,6,8)(3,7), (1,2,5,6), (1,2,5,6)(4,8), (1,2,5,6)(3,4)(7,8),
  (1,2,5,6)(3,4,7,8), (1,2,5,6)(3,7), (1,2,5,6)(3,7)(4,8),
  (1,2,5,6)(3,8,7,4), (1,2,5,6)(3,8)(4,7), (1,2,7,5,6,3),
  (1,2,7,5,6,3)(4,8), (1,2,7,8,5,6,3,4), (1,2,7,8)(3,4,5,6),
  (1,2,7)(3,5,6)(4,8), (1,2,7,4)(3,8,5,6), (1,2,7,4,5,6,3,8),
  (1,2,8,7,5,6,4,3), (1,2,8,3)(4,7,5,6), (1,2,8,5,6,4),
  (1,2,8,7)(3,5,6,4), (1,2,8,3,5,6,4,7), (1,2,8,5,6,4)(3,7),
  (1,2,8)(3,7)(4,5,6) ]
gap> Length(pHel);
38
gap> #The orders of each symmetry will be in PHel.
gap>  PHel:=pHel;;
gap> OrdH:=[];;
gap> for j in [1..Length(pHel)] do
>       OrdH[j]:=Order(pHel[j]);
> od;
gap> OrdH;
[ 4, 4, 6, 4, 8, 6, 6, 8, 4, 4, 8, 6, 8, 4, 6, 6, 4, 4, 4, 4, 4, 4,
  4, 4, 6, 6, 8, 4, 6, 4, 8, 8, 4, 6, 4, 8, 6, 6 ]
```

From the above symmetries, we'll classify those of order 4. And we'll see which ones are conjugate to obtain a representative. Similarly we will do the process for symmetries of order 6 and 8.

```
#First we will classify the previous symmetries of order 4.
gap> SimOrd4:=[];; #Aquí se guardarán los ciclos (S1) de orden 4.
gap> for j in [1..Length(pHel)] do
>       if Order(pHel[j])=4 then
>               Add(SimOrd4,pHel[j]);
>       fi;
> od;
gap> SimOrd4;
[ (1,2)(3,4,7,8)(5,6), (1,2)(3,8,7,4)(5,6), (1,2,3,4)(5,6,7,8),
  (1,2,3,8)(4,5,6,7), (1,2,4,3)(5,6,8,7), (1,2,4,7)(3,5,6,8),
  (1,2,5,6), (1,2,5,6)(4,8), (1,2,5,6)(3,4)(7,8), (1,2,5,6)(3,4,7,8),
  (1,2,5,6)(3,7), (1,2,5,6)(3,7)(4,8), (1,2,5,6)(3,8,7,4),
  (1,2,5,6)(3,8)(4,7), (1,2,7,8)(3,4,5,6), (1,2,7,4)(3,8,5,6),
  (1,2,8,3)(4,7,5,6), (1,2,8,7)(3,5,6,4) ]
gap> Length(SimOrd4);
18
gap> ElConj4:=[];;
gap>    for j in [2..Length(SimOrd4)] do
>               for k in [1..Length(eGC)] do
>                       if  eGC[k]*SimOrd4[1]*eGC[k]^-1= SimOrd4[j] then
>                               Add(ElConj4,
```

```
[j,Factorization(P,SimOrd4[j]),Factorization(P,eGC[k])]);
>                          else
>                                  continue;
>                          fi;
>                  od;
>          od;
gap> ElConj4;; #These are the conjugate symmetries of order 4.
```

A.1.2 Candidates to be S_2 of the the 16-cell

ElFigV returns the symmetry group of the vertex figure of 16-cell. These symmetries fix the base vertex. $v = \{1\}$.

The goal is find polyhedra with skew faces, so we search within the group of symmetries of the vertex figure of 16-cell, the candidate symmetries to construct a skew vertex figure are shown in red, because these are all the symmetries that are not involutions and send the 4 to 2 and those that do not construct skew cycles are discarded ((2,3,4)(6,7,8), (2,7,4)(3,8,6), (2,8,6,4), (2,8,6,4)(3,7)).

On the other hand, in the search for polyhedra with helical faces, all the symmetries that are not involutions and that send 8 to 2 are marked in blue. As we are looking to construct planar vertex figures, the permutations (2,3,4,6,7,8) and (2,7,4,6,3,8) are discarded since they form skew cycles.

```
gap> GFigVer:=Group(r1,r2,r3);
Group([ (2,3)(6,7), (3,4)(7,8), (4,8) ])
gap> ElFigV:=Elements(GFigVer);
```

[(), (4,8), (3,4)(7,8), (3,4,7,8), (3,7), (3,7)(4,8), (3,8,7,4), (3,8)(4,7), (2,3)(6,7), (2,3)(4,8)(6,7), (2,3,4)(6,7,8), (2,3,4,6,7,8), (2,3,6,7), (2,3,6,7)(4,8), (2,3,8,6,7,4), (2,3,8)(4,6,7), (2,4,3)(6,8,7), (2,4,7,6,8,3), (2,4)(6,8), (2,4,6,8), (2,4,3,6,8,7), (2,4,7)(3,6,8), (2,4)(3,7)(6,8), (2,4,6,8)(3,7), (2,6), (2,6)(4,8), (2,6)(3,4)(7,8), (2,6)(3,4,7,8), (2,6)(3,7), (2,6)(3,7)(4,8), (2,6)(3,8,7,4), (2,6)(3,8)(4,7), (2,7,6,3), (2,7,6,3)(4,8), (2,7,8,6,3,4), (2,7,8)(3,4,6), (2,7)(3,6), (2,7)(3,6)(4,8), (2,7,4)(3,8,6), (2,7,4,6,3,8), (2,8,7,6,4,3), (2,8,3)(4,7,6), (2,8,6,4), (2,8)(4,6), (2,8,7)(3,6,4), (2,8,3,6,4,7), (2,8,6,4)(3,7), (2,8)(3,7)(4,6)]

```
gap> Length(ElFigV);
48
```

A.1.3 A regular polyhedron with helical faces of the 16-cell

```
gap> s1:=r0*r1*r2*r3;
(1,8,7,6,5,4,3,2)
gap> s2:=r3*r2*r1*r2;
(2,4,6,8)
```

Calculating the right transversals of $\langle S_1 \rangle$, $\langle S_1 S_2 \rangle$ and $\langle S_2 \rangle$ in $G = \langle S_1, S_2 \rangle$.

```
gap> G:=Group(s1,s2);
Group([ (1,8,7,6,5,4,3,2), (2,4,6,8) ])
gap> Size(G);
32
gap> genS1:=Group(s1);
Group([ (1,8,7,6,5,4,3,2) ])
gap> Elements(RightTransversal(G,genS1));
[ (), (2,4,6,8), (2,6)(4,8), (2,8,6,4) ]
gap> genS1S2:=Group(s1*s2);
Group([ (1,2)(3,4)(5,6)(7,8) ])
gap> Elements(RightTransversal(G,genS1S2));
[ (), (2,4,6,8), (2,6)(4,8), (2,8,6,4), (1,2,3,4,5,6,7,8),
  (1,2,5,6)(3,4,7,8), (1,2,7,8,5,6,3,4), (1,3,5,7)(2,4,6,8),
```

```
  (1,3,5,7)(2,6)(4,8), (1,3,5,7)(2,8,6,4), (1,4,7,2,5,8,3,6),
  (1,4)(2,7)(3,6)(5,8), (1,5)(2,6)(3,7)(4,8), (1,5)(2,8,6,4)(3,7),
  (1,6,3,8,5,2,7,4), (1,7,5,3)(2,8,6,4) ]
gap> genS2:=Group(s2);
Group([ (2,4,6,8) ])
gap> Elements(RightTransversal(G,genS2));
[ (), (1,2,3,4,5,6,7,8), (1,3,5,7)(2,4,6,8), (1,4,7,2,5,8,3,6),
  (1,5)(2,6)(3,7)(4,8), (1,6,3,8,5,2,7,4), (1,7,5,3)(2,8,6,4),
  (1,8,7,6,5,4,3,2) ]
```

A.2 8-cell and 24-cell

A.2.1 Cycles of the 24-cell

Distinguished generators of the 24-cell:

```
gap> r0:=(1,4)(2,3)(5,8)(6,7)(9,12)(10,11)(13,16)(14,15)(19,23);
(1,4)(2,3)(5,8)(6,7)(9,12)(10,11)(13,16)(14,15)(19,23)
gap> r1:=(2,24)(4,21)(5,19)(7,18)(8,14)(10,17)(12,20)(13,22)(15,23);
(2,24)(4,21)(5,19)(7,18)(8,14)(10,17)(12,20)(13,22)(15,23)
gap> r2:=(5,13)(6,14)(7,15)(8,16)(17,20)(21,24);
(5,13)(6,14)(7,15)(8,16)(17,20)(21,24)
gap> r3:=(2,13)(3,16)(6,9)(7,12)(18,20)(22,24);
(2,13)(3,16)(6,9)(7,12)(18,20)(22,24)
gap> P:=Group(r0,r1,r2,r3);
<permutation group with 4 generators>
gap> eGC:=Elements(P);;
gap> v:=11;
11
gap> e:=10;
10
gap> #We will remove symmetries of order 2 and 3.
gap> filP:=[];; ##Filtro de P.
gap> for j in [1..Length(eGC)] do
>       if (Order(eGC[j])=2 or Order(eGC[j])=3)=true then
>               continue;
>       else
>               Add(filP,eGC[j]);
>       fi;
> od;
gap> filP;;
gap> Length(filP);
933
gap> #The base vertex is 11, because it is a fixed vertex for the three reflections: r1, r2
gap> #The base vertex is 11 and the base edge is {11, 10}, then we want a rotation that sen
when it rotates for the first time.
gap> pHel:=[];; #Possible S1 cycles will be saved here.
gap> for i in [1..Length(filP)] do
>       if Length(Cycle(filP[i],v))>1 and Cycle(filP[i],v)[2]=e
                and Length(Cycle(filP[i],v))=Order(filP[i]) then
>               Add(pHel,filP[i]);
>       fi;
> od;
gap> pHel;
```

```
[ (1,4,3,2)(5,8,7,6)(9,12,11,10)(13,16,15,14)(18,19,22,23),
  (1,4,3,2)(5,16,7,14)(6,13,8,15)(9,12,11,10)(17,20)(18,19,22,23)(21,24),
  (1,4,8,7,6,2)(3,5)(9,15)(10,14,13,16,12,11)(17,18,19,21,22,23),
  (1,4,8,12,11,10,14,2)(3,5,16,7,9,15,6,13)(17,20,18,19,21,24,22,23),
  (1,4,16,12,11,10,6,2)(3,13,8,15,9,7,14,5)(17,18,19,24,21,22,23,20),
  (1,4,16,15,14,2)(3,13)(5,8,12,11,10,6)(7,9)(18,19,24,22,23,20),
  (1,4,8,5)(2,3,7,6)(9,13,16,12)(10,14,15,11)(17,19,21,23),
  (1,4,8,12,9,13)(2,3,7,11,10,14)(5,16)(6,15)(17,20,19,21,24,23),
  (1,4,16,12,9,5)(2,3,15,11,10,6)(7,14)(8,13)(17,19,24,21,23,20),
  (1,4,16,13)(2,3,15,14)(5,8,12,9)(6,7,11,10)(19,24,23,20),
  (1,4,3,7,6,5)(2,8)(9,13,16,15,11,10)(12,14)(17,19,22,21,23,18),
  (1,4,3,7,11,10,9,13)(2,8,15,6,12,14,5,16)(17,20,19,22,21,24,23,18),
  (1,4,16,15,11,10,6,5)(2,8,13,3,12,14,7,9)(17,19,24,22,21,23,20,18),
  (1,4,16,13)(2,8,15,9)(3,12,14,5)(6,7,11,10)(17,18)(19,24,23,20)(21,22),
  (1,4,8,5)(2,16,7,9)(3,12,6,13)(10,14,15,11)(17,19,21,23)(18,20)(22,24),
  (1,4,8,7,11,10,14,13)(2,16,5,3,12,6,15,9)(17,18,20,19,21,22,24,23),
  (1,4,3,15,11,10,9,5)(2,16,7,14,12,6,13,8)(17,19,22,24,21,23,18,20),
  (1,4,3,15,14,13)(2,16)(5,8,7,11,10,9)(6,12)(18,20,19,22,24,23),
  (1,4,24,3,2,21)(5,23,6,15,19,16)(7,18,14,13,22,8)(9,12,17,11,10,20),
  (1,4,22,8,12,17,11,10,18,14,2,21)(3,13,24,16,5,23,9,7,20,6,15,19),
  (1,4,22,8,5,24)(2,21,16,12,17,6)(3,13,23,9,7,19)(10,18,14,15,20,11),
  (1,4,23,11,10,19)(2,21,3,15,18,14)(5,22,8,12,17,9)(6,13,24,16,7,20),
  (1,4,23,11,10,19)(2,21,16,12,17,6)(3,15,20,9,5,24)(7,18,14,13,22,8),
  (1,4,24,8,7,17,11,10,20,14,13,21)(2,22,3,5,23,6,12,18,9,15,19,16),
  (1,4,23,11,10,19)(2,22,3,12,18,9)(5,21,8,15,17,14)(6,13,24,16,7,20),
  (1,4,23,11,10,19)(2,22,3,12,18,9)(5,24,8,7,17,6)(13,21,16,15,20,14),
  (1,4,21,16,15,20,11,10,17,6,5,24)(2,22,3,13,23,14,12,18,9,7,19,8),
  (1,4,24,3,7,18,11,10,20,9,13,22)(2,23,6,15,17,14,12,19,16,5,21,8),
  (1,4,24,8,5,22)(2,23,6,12,19,16)(3,7,17,9,13,21)(10,20,14,15,18,11),
  (1,4,21,3,15,18,11,10,17,9,5,22)(2,23,14,7,20,6,12,19,8,13,24,16),
  (1,4,21,16,13,22)(2,23,14,12,19,8)(3,15,20,9,5,24)(6,7,18,11,10,17),
  (1,4,21,3,2,24)(5,22,16,15,18,6)(7,19,8,13,23,14)(9,12,20,11,10,17),
  (1,4,22,16,12,20,11,10,18,6,2,24)(3,5,21,8,13,23,9,15,17,14,7,19),
  (1,4,22,16,13,21)(2,24,8,12,20,14)(3,5,23,9,15,19)(6,7,17,11,10,18),
  (1,4,23,11,10,19)(2,24,3,7,18,6)(5,21,8,15,17,14)(9,13,22,16,12,20),
  (1,4,23,11,10,19)(2,24,8,12,20,14)(3,7,17,9,13,21)(5,22,16,15,18,6) ]
gap> Length(pHel);
36
gap> #The orders of each symmetry will be in PHel.
gap> PHel:=pHel;;
gap> OrdH:=[];;
gap> for j in [1..Length(pHel)] do
>       OrdH[j]:=Order(pHel[j]);
> od;
gap> OrdH;
[ 4, 4, 6, 8, 8, 6, 4, 6, 6, 4, 6, 8, 8, 4, 4, 8, 8, 6, 6, 12, 6, 6,
  6, 12, 6, 6, 12, 12, 6, 12, 6, 6, 12, 6, 6, 6 ]
```

Note: The programming for classifying conjugate symmetries is analogous to that shown for the 16-cell case.

A.3 120-cell

A.3.1 Cycles of the 120-celll

The base vertex and distinguished generators of the 120-cell are based in the article [2] and appear in [1]. These are the following:

```
gap> rho := (1 + Sqrt(5)) / 2;; #Golden ratio.
gap> #Distinguished generators of the 120-cell as reflection matrices.
gap> r0 := [ [1, 0, 0, 0],
>          [0, 1, 0, 0],
>          [0, 0, -1, 0],
>          [0, 0, 0, 1] ];;
gap> r1 := (1/2) * [ [2, 0, 0, 0],
>                    [0, rho, -1, rho - 1],
>                    [0, -1, 1 - rho, rho],
>                    [0, rho - 1, rho, 1] ];;
gap> r2 := [ [1, 0, 0, 0],
>          [0, 1, 0, 0],
>          [0, 0, 1, 0],
>          [0, 0, 0, -1] ];;
gap> r3 := (1/2) * [ [rho, 1, 0, -1 + rho],
>                    [1, 1 - rho, 0, -rho],
>                    [0, 0, 2, 0],
>                    [-1 + rho, -rho, 0, 1] ];;
gap> v:=[1+rho,1,-2+rho,0];; #Base vertex.
gap> P:=Group(r0,r1,r2,r3);;
gap> eGC:=Elements(P);;
gap> e:=v*r0;;
gap> OrdC:=[];;
gap> for j in [1..Length(eGC)] do
>        if Order(eGC[j])>1 then
>                Add(OrdC,Order(eGC[j]));
>        fi;
> od;
gap> Set(OrdC); #Length of the cycles.
[ 2, 3, 4, 5, 6, 10, 12, 15, 20, 30 ]
gap> # We will save the cycles that contain v and e (base vertex and edge)
# in Ciclos.
gap> Ciclos:=[];;
gap> Simgc:=[];; # In Simgc we will save the candidate symmetries to be S1.
#We will remove symmetries of order 3 since they correspond to triangular
#faces of the 120-cell.
gap> for j in [1..Length(eGC)] do
>        ciclo:=[];
>        for i in [1..Order(eGC[j])] do
>                Add(ciclo,v*eGC[j]^i);
>        od;
>        if Length(Set(ciclo))>3 and v*eGC[j]^-1=e and
 Length(Set(ciclo))=Order(eGC[j]) then
>                Add(Ciclos,ciclo);
>                Add(Simgc,eGC[j]);
>        fi;
> od;
gap> OCiclos:=[];;
```

```
gap> for j in [1..Length(Ciclos)] do
>       Add(OCiclos,Length(Ciclos[j]));
> od;
gap> Set(OCiclos); # Length of cycles containing v and e.
[ 5, 10, 30 ]
```

Classifyng symmetries with respect to their order. We will use symmetries of order 4 as an example and then obtain their conjugation class to obtain a representative:

```
gap> OSim:=5;;
gap> SimOrd:=[];; #Candidate to be S1 of order 5.
gap> for j in [1..Length(Simgc)] do
>       if Order(Simgc[j])=OSim then
>               Add(SimOrd,Simgc[j]);
>       fi;
> od;
gap> SimOrd;;
gap> Length(SimOrd);
3
gap> Conj:=[];;
gap>    for j in [2..Length(SimOrd)] do
>               for k in [1..Length(eGC)] do
>                       if eGC[k]*SimOrd[1]*eGC[k]^-1= SimOrd[j] then
>                               Add(Conj,[j,Factorization(P,SimOrd[j]),
Factorization(P,eGC[k])]);
>                       else
>                               continue;
>                       fi;
>               od;
>       od;
gap> Conj;; #This list contains the elements of the conjugation class of
#SimOrd[1].
gap> # We chose as representative element number 2 from the previous list,
#which is:
gap> Factorization(P,SimOrd[]);
x1*x2
gap> #So the candidate to be S1 of order 5 is the following:
gap> s1:=R0*R1;;
```

We will obtain the space in which are the vertices of the cycle, for which we will write the following function called espacio with parameters a symmetry S and a vertex v.

```
gap> espacio:= function(S,v)
> A:=[];
> for i in [0..Order(S)-1] do
>       Add(A,v*S^i);
> od;
> B:=[];
> for i in [1..Order(S)] do
>       Add(B,A[i]-v);
> od;
> return(RankMat(B));
> end;
function( S, v ) ... end

gap> s1:=R0*R1;; #Order 5.
gap> espacio(s1,v); # Plane (This correspond to a pentagonal face of the 120-cell).
```

```
2
```

Similarly we classify the elements according to their order and choose a representative of each conjugation class.

```
gap> s1:=R0*R1*R2;; #Order 10.
gap> espacio(s1,v); #Skew.
3

gap> s1:=R0*R1*R2*R3;; #Order 30.
gap> espacio(s1,v); #Helix.
4
```

A.3.2 Candidates to be S_2 of the 120-cell for S_1 skews

$\underline{S_1 = R_0R_1R_2.}$

```
gap> GVerFig:=Group(r1,r2,r3); #Group of the vertex figure of 120-cell
<matrix group with 3 generators>
gap> ElVF:=Elements(GVerFig);;
gap> #Defining S1 as A1.
gap> S1:=r0*r1*r2;; #Order 10.
gap> #The candidates for S2 are those that are in the group of the 120-cell vertex
 #figure, we will remove the identity and the elements of order 2 and 3, because
 #these symmetries would represent digons or would be in a plane, respectively.
gap> CandSigma2:=[];; #Candidates to be S2
gap> for j in [1..Length(ElVF)] do
>       if (Order(ElVF[j])=1 or Order(ElVF[j])=2 or Order(ElVF[j])=3)=true then
>               continue;
>       else
>               Add(CandSigma2,ElVF[j]);
>       fi;
> od;
gap> CandSigma2;;
gap> Length(CandSigma2);
6
gap> #Of the possible candidates to be S2 we will discard those that do not satisfy the fol
gap> S2:=[];;
gap> for j in [1..Length(CandSigma2)] do
>       if Order(S1*CandSigma2[j])=2 then
>               Add(S2,Factorization(P,CandSigma2[j]));
>       fi;
> od;
gap> S2;
[ x2*x3*x4*x2*x3, x3*x4*x2 ]

gap> u:=v*S1;;

gap> S2:=R1*R2*R3*R1*R2;;
gap> Order(S2);
4
gap> espacio(S2,u); #Skew.
3

gap> S2:=R2*R3*R1;;
```

```
gap> Order(S2);
4
gap> espacio(S2,u); #Skew.
3
```

A.3.3 Candidates to be S_2 of the 120-cell for S_1 helical

$S_1 = R_0R_1R_2R_3$.

```
gap> S1:=r0*r1*r2*r3;; #Order 30.
gap> CandSigma2:=[];; #Candidates to be S2
gap> for j in [1..Length(ElVF)] do
>       if (Order(ElVF[j])=1 or Order(ElVF[j])=2)=true then
>               continue;
>       else
>               Add(CandSigma2,ElVF[j]);
>       fi;
> od;
gap> CandSigma2;;
gap> Length(CandSigma2);
14
gap> #Of the possible candidates to be S2 we will discard those that do not satisfy
# the following condition: Order(S1*S2)=2;
gap> S2:=[];;
gap> for j in [1..Length(CandSigma2)] do
>       if Order(S1*CandSigma2[j])=2 then
>               Add(S2,Factorization(P,CandSigma2[j]));
>       fi;
> od;
gap> S2;
[ x4*x2*x3*x2, x3*x4*x3*x2, x4*x3*x2 ]

gap> u:=v*S1;;

gap> S2:=r3*r1*r2*r1;;
gap> Order(S2);
3
gap> espacio(S2,u); #Plane.
2

gap> S2:=r2*r3*r2*r1;;
gap> Order(S2);
3
gap> espacio(S2,u);  #Plane.
2

gap> S2:=r3*r2*r1;;
gap> Order(S2);
4
gap> espacio(S2,u); #Skew.
3
```

Searching for the polyhedron $\{3,30\}$ in the list [3].

Group of the first type polyhedron $\{30,3\}$ which appears in [3] listed as :
R97.9 : Type $\{3,30\}_{20}$ of order 2880. We will call L the group of this polyhedron.

```
gap> l:= FreeGroup( "l0", "l1","l2" );;
gap> L:=l/[ l.1^2, l.2^-3, (l.2 * l.3)^2, (l.2 * l.1)^2, (l.3 * l.1)^2,
l.3 * l.2 * l.3^-2 * l.2 * l.3^-2 * l.2 * l.3^-1 * l.2 * l.3^-2 * l.2
* l.3^2 * l.2^-1 * l.3^2, l.3 * l.2 * l.3^-3 * l.2^-1
* l.3^5 * l.2^-1 * l.3^-3 * l.2 * l.3^6 ];
<fp group on the generators [ l0, l1, l2 ]>
gap> Size(L);
2880
gap> cc:=ConjugacyClassesSubgroups(L);;
rs1:=[];; #In rs1 we will save the subgroups of L of order 1440.
gap> for i in [1..Length(cc)] do
>       if Size(Representative(cc[i]))=1440 then
>               Add(rs1,Representative(cc[i]));
>       fi;
> od;
gap> Length(rs1); #There are three subgroups of L of order 1440.
3
```

Group of the second polyhedron of type $\{30,3\}$ which appears in [3] listed as:
R97.10 : Type $\{3,30\}_{40}$ of order 2880. We will name F the group of this polyhedron.

```
gap> #R97.10 :  Type {3,30}_40  Order 2880  mV = 2  mF = 1
gap> f:= FreeGroup( "f0", "f1","f2" );;
gap> F:=f/[ f.3^2, f.1^-3, (f.1 * f.2)^2, (f.1 * f.3)^2, (f.2 * f.3)^2,
 (f.2^-2 * f.1 * f.2^-1)^3 ];
<fp group on the generators [ f0, f1, f2 ]>
gap> Size(F);
2880
gap> ccs:=ConjugacyClassesSubgroups(F);;
gap> rs2:=[];; #In rs2 se will save the subgroups of F of order 1440.
gap> for i in [1..Length(ccs)] do
>       if Size(Representative(ccs[i]))=1440 then
>               Add(rs2,Representative(ccs[i]));
>       fi;
> od;
gap> Length(rs2); #There is only one subgroup of F of order 1440.
1
```

G is the symmetry group of the 120-cell.

```
gap> g:=FreeGroup("r0","r1","r2","r3");
<free group on the generators [ r0, r1, r2, r3 ]>
gap> G:=g/[g.1^2,g.2^2,g.3^2,g.4^2,(g.1*g.2)^5,(g.2*g.3)^3,(g.3*g.4)^3,
(g.1*g.3)^2,(g.1*g.4)^2,(g.2*g.4)^2];
<fp group on the generators [ r0, r1, r2, r3 ]>
gap> Order(G);
14400
gap> r0:=G.1;;
gap> r1:=G.2;;
gap> r2:=G.3;;
gap> r3:=G.4;;
```

We then define S_1 and S_2 and the symmetry group of the polyhedron $\mathsf{P}(\mathsf{S}_1,\mathsf{S}_2;\mathsf{v})$ will be grup303.

```
gap> s1:=r0*r1*r2*r3;
r0*r1*r2*r3
gap> s2:=r3*r2*r1*r3;
r3*r2*r1*r3
gap> grup303:=Group(s1,s2);
Group([ r0*r1*r2*r3, r3*r2*r1*r3 ])
```

Now we will see if any of the subgroups of L (R97.9) of order 1440 is isomorphic to the symmetry group of the polyhedron $\mathsf{P}(\mathsf{S}_1, \mathsf{S}_2)$.

```
gap> LoadPackage("sonata");
gap> IsIsomorphicGroup(rs1[1],grup303);
false
gap> IsIsomorphicGroup(rs1[2],grup303);
false
gap> IsIsomorphicGroup(rs1[3],grup303);
false
```

Now, we will now see if the only subgroup of F (R97.10) of order 1440 is isomorphic to the group of symmetries of the polyhedron $\mathsf{P}(\mathsf{S}_1, \mathsf{S}_2)$.

```
gap> IsIsomorphicGroup(rs2[1],grup303);
true
```

Therefore the subgroup of rotations of F (R97.10) is isomorphic to the group of $\mathsf{P}(S_1, S_2)$.

A.4 600-cell

A.4.1 Cycles of the 600-cell

The base vertex and distinguished generators of the 600-cell are based in the article [2] and appear in [1]. These are the following:

```
gap> rho := (1 + Sqrt(5)) / 2;; #Golden ratio.
gap> #Distinguished generators of the 600-cell as reflection matrices.
gap> R0 := (1/2) * [ [rho, 1, 0, -1 + rho],
>                    [1, 1 - rho, 0, -rho],
>                    [0, 0, 2, 0],
>                    [-1 + rho, -rho, 0, 1] ];;
gap>
gap> R1 := [ [1, 0, 0, 0],
>          [0, 1, 0, 0],
>          [0, 0, 1, 0],
>          [0, 0, 0, -1] ];;
gap>
gap> R2 := (1/2) * [ [2, 0, 0, 0],
>                    [0, rho, -1, rho - 1],
>                    [0, -1, 1 - rho, rho],
>                    [0, rho - 1, rho, 1] ];;
gap>
gap> R3 := [ [1, 0, 0, 0],
>          [0, 1, 0, 0],
>          [0, 0, -1, 0],
>          [0, 0, 0, 1] ];;
gap> v:=[1,0,0,0];; #Base vertex.
gap> P:=Group(R0,R1,R2,R3);;
```

```
gap> eGC:=Elements(P);;
gap> e:=v*R0;; # Base edge.
gap> OrdC:=[];;
gap> for j in [1..Length(eGC)] do
>       if Order(eGC[j])>1 then
>               Add(OrdC,Order(eGC[j]));
>       fi;
> od;
gap> Set(OrdC); # Length of the cycles.
[ 2, 3, 4, 5, 6, 10, 12, 15, 20, 30 ]
gap> # We will save the cycles that contain v and e (base vertex and edge)
# in Ciclos.
gap> Ciclos:=[];;
gap> Simgc:=[];; # In Simgc we will save the candidate symmetries to be S1.
#We will remove symmetries of order 3 since they correspond to triangular
#faces of the 600-cell.
gap> for j in [1..Length(eGC)] do
>       ciclo:=[];
>       for i in [1..Order(eGC[j])] do
>               Add(ciclo,v*eGC[j]^i);
>       od;
>       if Length(Set(ciclo))>3 and v*eGC[j]^-1=e and
                Length(Set(ciclo))=Order(eGC[j]) then
>               Add(Ciclos,ciclo);
>               Add(Simgc,eGC[j]);
>       fi;
> od;
gap> OCiclos:=[];;
gap> for j in [1..Length(Ciclos)] do
>       Add(OCiclos,Length(Ciclos[j]));
> od;
gap> Set(OCiclos); # Length of cycles containing v and e.
[ 4, 5, 6, 10, 12, 15, 20, 30 ]
gap> #Classifyng symmetries with respect to their order:
gap> #We will use symmetries of order 4 as an example and then obtain their
#conjugation class to obtain a representative:
gap> OSim:=4;;
gap> SimOrd:=[];; #Candidate to be S1 of order 4.
gap> for j in [1..Length(Simgc)] do
>       if Order(Simgc[j])=OSim then
>               Add(SimOrd,Simgc[j]);
>       fi;
> od;
gap> SimOrd;;
gap> Length(SimOrd);
10
gap> Conj:=[];;
gap>    for j in [2..Length(SimOrd)] do
>               for k in [1..Length(eGC)] do
>                       if  eGC[k]*SimOrd[1]*eGC[k]^-1= SimOrd[j] then
>                               Add(Conj,[j,Factorization(P,SimOrd[j]),
Factorization(P,eGC[k])]);
>                       else
>                               continue;
```

```
>                           fi;
>                   od;
>           od;
gap>Conj;; #This list contains the elements of the conjugation class of
#SimOrd[1].
gap> # We chose as representative element number 7 from the previous list,
#which is:
gap> Factorization(P,SimOrd[7]);
x1*x2*x3
gap> #So the candidate to be S1 of order 4 is the following:
gap> s1:=R0*R1*R2;; #Orden 4
```

We will obtain the space in which are the vertices of the cycle, for which we will write the following function called espacio with parameters a symmetry S and a vertex v.

```
gap> espacio:= function(S,v)
> A:=[];
> for i in [0..Order(S)-1] do
>           Add(A,v*S^i);
> od;
> B:=[];
> for i in [1..Order(S)] do
>           Add(B,A[i]-v);
> od;
> return(RankMat(B));
> end;
function( S, v ) ... end

gap> s1:=R0*R1*R2;; #Orden 4
gap> espacio(s1,v); # Skew (This correspond to the Petrie on the base tetrahedron).
3
```

Similarly we classify the elements according to their order and choose a representative of each conjugation class.

```
gap> s1:=R0*R1*R2*R3*R2*R1;; #Order 5
gap> espacio(s1,v); #Plane.
2

gap> s1:=R3*R0*R1*R2*R3*R2*R1;; #Order 6
gap> espacio(s1,v); #Skew.
3

gap> #There are four conjugacy classes for symmetries of order 10. Their
 #respective representatives are s1, sa, sb and sc.

gap> s1:=R0*R1*R2*R3*R1;; #Order 10
gap> A:=[];;
gap> for i in [0..Order(s1)-1] do
>           Add(A,v*s1^i);
> od;
gap> RankMat(A);
4
gap> B:=[];;
gap> for i in [1..Order(s1)] do
>           Add(B,A[i]-v);
> od;
```

```
gap> RankMat(B); #Skew.
3
gap> sa:=R3*R0*(R1*R2*R3)^3*R1*R2*R1;;
gap> A1:=[];;
gap> for i in [0..Order(s1)-1] do
>       Add(A1,v*sa^i);
> od;
gap> RankMat(A1);
2
gap> B1:=[];;
gap> for i in [1..Order(s1)] do
>       Add(B1,A1[i]-v);
> od;
gap> RankMat(B1); #Plane.
2
gap> sb:=R0*(R1*R2*R3)^3*R2*R1;;
gap> B1:=[];;
gap> for i in [0..Order(s1)-1] do
>       Add(B1,v*sb^i);
> od;
gap> sc:=R3*R2*R3*R0*(R1*R2*R3)^3*R1*R2*R1;;
gap> C1:=[];;
gap> for i in [0..Order(s1)-1] do
>       Add(C1,v*sc^i);
> od;
gap> Set(A)=Set(A1);
false
gap> Set(A1)=Set(B1);
true
gap> Set(B1)=Set(C1);
true

gap> #The cycle generated by the representative of the first conjugacy class is
  #skew, while the cycles generated by the respective representatives of the
  #following three conjugacy classes contain the same vertices and those are in
  #a plane.

gap> s1:=R0*(R1*R2*R3)^3;; #Order 12
gap> espacio(s1,v); #Helix.
4

gap> s1:=R2*R3*R0*R1*R2*R3*R2*R1;; #Order 15
gap> espacio(s1,v); #Helix.
4

gap> s1:=R2*R3*R0*R1*R2*R3;; #Order 20
gap> espacio(s1,v); #Helix.
4

gap> s1:=R0*R1*R2*R3;; #Order 30
gap> espacio(s1,v); #Helix.
4
```

A.4.2 Candidates to be S_2 of the 600-cell for S_1 skews

$S_1 = R_0R_1R_2$.

```
gap> GVerFig:=Group(R1,R2,R3); #Group of the vertex figure of 600-cell
<matrix group with 3 generators>
gap> ElVF:=Elements(GVerFig);;
gap> #Defining S1 as S1.
gap> S1:=R0*R1*R2;; #Order 4.
gap> #The candidates for S2 are those that are in the group of the 600-cell
  #vertex figure, we will remove the identity and the elements of order 2 and 3,
  #because these symmetries would represent digons or would be in a plane,
  #respectively.
gap> CandSigma2:=[];; #Candidates to be S2
gap> for j in [1..Length(ElVF)] do
>       if (Order(ElVF[j])=1 or Order(ElVF[j])=2 or Order(ElVF[j])=3)=true then
>               continue;
>       else
>               Add(CandSigma2,ElVF[j]);
>       fi;
> od;
gap> CandSigma2;;
gap> Length(CandSigma2);
68
gap> #Of the possible candidates to be S2 we will discard those that do not
 #satisfy that the order of S1*S2 is 2.
gap> S2:=[];;
gap> for j in [1..Length(CandSigma2)] do
>       if Order(S1*CandSigma2[j])=2 then
>               Add(S2,Factorization(P,CandSigma2[j]));
>       fi;
> od;
gap> #The following symmetries are the candidates to be S2 for S1=R0*R1*R2.
gap> S2;
[ (x2*x3*x4)^2*x3, x3*x4*x2, x3*x4*x2*x3*x4, x2*x3*x4*x2*x3 ]

gap> u:=v*S1;;

gap> S2:=(R1*R2*R3)^2*R2;;
gap> Order(S2);
6
gap> espacio(S2,u); #Skew.
3

gap> S2:=R2*R3*R1;;
gap> Order(S2);
10
gap> espacio(S2,u); #Skew.
3

gap> S2:=R2*R3*R1*R2*R3;;
gap> Order(S2);
6
gap> espacio(S2,u); #Skew.
3
```

```
gap> S2:=R1*R2*R3*R1*R2;;
gap> Order(S2);
10
gap> espacio(S2,u); #Skew.
3
```

$\underline{S_1 = R_3R_0R_1R_2R_3R_2R_1.}$
We repeat the same code above to obtain the candidates to be S_2 .

```
S1:=R3*R0*R1*R2*R3*R2*R1;; #Orden 6.
#The following symmetries are the candidates to be S2 for S1=R3*R0*R1*R2*R3*R2*R1.
gap> S2;
[ x2*(x3*x4)^2*x2*x3*x4*x3, x2*(x3*x4)^2*x2*x3, x2*(x3*x4)^2*x2, (x2*x3*x4)^3,
  (x2*x3*x4)^2*x2 ]

gap> u:=v*S1;;

gap> S2:=R1*(R2*R3)^2*R1*R2*R3*R2;;
gap> Order(S2);
10
gap> espacio(S2,u); #Skew.
3

gap> S2:=R1*(R2*R3)^2*R1*R2;;
gap> Order(S2);
6
gap> espacio(S2,u); #Skew.
3

gap> S2:=R1*(R2*R3)^2*R1;;
gap> Order(S2);
5
gap> espacio(S2,u); #Plane.
2

gap> S2:=(R1*R2*R3)^3;;
gap> Order(S2);
10
gap> espacio(S2,u); #Skew.
3

gap> S2:=(R1*R2*R3)^2*R1;;
gap> Order(S2);
6
gap> espacio(S2,u); #Skew.
3
```

$\underline{S_1 = R_0R_1R_2R_3R_1.}$
We repeat the same code above to obtain the candidates to be S_2 .

```
gap> S1:=R0*R1*R2*R3*R1;; #Orden 10 skew.
gap> S2;
[ x4*x3*x4*x2*x3*x4*x3, x4*x2*x3*x2, x4*x3*x2, (x4*x2*x3)^2*x4,
```

```
  x4*x3*x4*x2*x3 ]

gap> u:=v*S1;;

gap> S2:=R3*R2*R3*R1*R2*R3*R2;;
gap> Order(S2);
6
gap> espacio(S2,u); #Skew
3

gap> S2:=R3*R1*R2*R1;;
gap> Order(S2);
5
gap> espacio(S2,u); #Plane
2

gap> S2:=R3*R2*R1;;
gap> Order(S2);
10
gap> espacio(S2,u); #Skew
3

gap> S2:=(R3*R1*R2)^2*R3;;
gap> Order(S2);
10
gap> espacio(S2,u); #Skew
3

gap> S2:=R3*R2*R3*R1*R2;;
gap> Order(S2);
6
gap> espacio(S2,u); #Skew
3
```

A.4.3 Candidates to be S_2 of the 600-cell for S_1 helical

$S_1 = R_0(R_1R_2R_3)^3$.

```
gap> S1:=R0*(R1*R2*R3)^3;; #Orden 12 #Helical
gap> CandSigma2:=[];; #Candidates to be S2
gap> for j in [1..Length(ElVF)] do
>       if (Order(ElVF[j])=1 or Order(ElVF[j])=2)=true then
>               continue;
>       else
>               Add(CandSigma2,ElVF[j]);
>       fi;
> od;
gap> CandSigma2;;
gap> Length(CandSigma2);
88
gap> #Of the possible candidates to be S2 we will discard those that do not satisfy the fol
gap> S2:=[];;
gap> for j in [1..Length(CandSigma2)] do
```

```
>         if Order(S1*CandSigma2[j])=2 then
>                 Add(S2,Factorization(P,CandSigma2[j]));
>         fi;
> od;
gap> S2;
[ x4*x3*x4*x2*x3*x4*x3*x2, x4*(x3*x4*x2)^3, (x4*x2*x3)^3*x2, x4*(x3*x4*x2)^2*x3*x2,
  (x4*x2*x3)^4 ]

gap> u:=v*S1;;

gap> S2:=R3*R2*R3*R1*R2*R3*R2*R1;;
gap> Order(S2);
3
gap> espacio(S2,u); #Plane.
2

gap> S2:=R3*(R2*R3*R1)^3;;
gap> Order(S2);
5
gap> espacio(S2,u); #Plane.
2

gap> S2:=(R3*R1*R2)^3*R1;;
gap> Order(S2);
3
gap> espacio(S2,u);  #Plane.
2

gap> S2:=R3*(R2*R3*R1)^2*R2*R1;;
gap> Order(S2);
10
gap> espacio(S2,u); #Skew.
3

gap> S2:=(R3*R1*R2)^4;;
gap> Order(S2);
5
gap> espacio(S2,u); #Plane.
2
```

$\underline{S_1 = R_2R_3R_0R_1R_2R_3R_2R_1.}$

```
gap> S1:=R2*R3*R0*R1*R2*R3*R2*R1;;#Orden 15
gap> #We repeat the same code above to obtain the candidates to be $S_2$ .
gap> S2;
[ (x2*x3*x4)^3*x3, x2*(x3*x4)^2*x2*x3, x2*(x3*x4)^2*x2*x3*x4, x2*(x3*x4)^2*x2,
  (x2*x3*x4)^2*x2*x3 ]

gap> u:=v*S1;;

gap> S2:=(R1*R2*R3)^3*R2;;
gap> Order(S2);
3
gap> espacio(S2,u); #Plane
```

```
2

gap> S2:=R1*(R2*R3)^2*R1*R2;;
gap> Order(S2);
6
gap> espacio(S2,u); #3 Skew.
3

gap> S2:=R1*(R2*R3)^2*R1*R2*R3;;
gap> Order(S2);
3
gap> espacio(S2,u); #Plane.
2

gap> S2:=R1*(R2*R3)^2*R1;;
gap> Order(S2);
5
gap> espacio(S2,u); #Plane.
2

gap> S2:=(R1*R2*R3)^2*R1*R2;;
gap> Order(S2);
5
gap> espacio(S2,u); #Plane.
2
```

$\underline{S_1 = R_2R_3R_0R_1R_2R_3.}$

```
gap> S1:=R2*R3*R0*R1*R2*R3;; #Order 20.
gap> S2;
[ (x4*x2*x3)^2*x4*x3, x4*x2*x3*x2, x4*x3*x4*x2, (x4*x2*x3)^2, x4*x3*x4*x2*x3 ]

gap> u:=v*S1;;

gap> S2:=(R3*R1*R2)^2*R3*R2;;
gap> Order(S2);
3
gap> espacio(S2,u); #Plane.
2

gap> S2:=R3*R1*R2*R1;;
gap> Order(S2);
5
gap> espacio(S2,u); #Plane.
2

gap> S2:=R3*R2*R3*R1;;
gap> Order(S2);
3
gap> espacio(S2,u); #Plane.
2

gap> S2:=(R3*R1*R2)^2;;
gap> Order(S2);
```

```
5
gap> espacio(S2,u); #Plane.
2

gap> S2:=R3*R2*R3*R1*R2;;
gap> Order(S2);
6
gap> espacio(S2,u); #Skew.
3
```

$\underline{S_1 = R_0R_1R_2R_3.}$

```
gap> S1:=R0*R1*R2*R3;; #Orden 30
gap> S2;
[ (x4*x2*x3)^2*x4*x3, x4*x2*x3*x2, x4*x3*x2, x4*x3*x4*x2, (x4*x2*x3)^2 ]

gap> u:=v*S1;;

gap> S2:=(R3*R1*R2)^2*R3*R2;;
gap> Order(S2);
3
gap> espacio(S2,u); #Plane.
2

gap> S2:=R3*R1*R2*R1;;
gap> Order(S2);
5
gap> espacio(S2,u); #Plane.
2

gap> S2:=R3*R2*R1;;
gap> Order(S2);
10
gap> espacio(S2,u); #Skew.
3

gap> S2:=R3*R2*R3*R1;;
gap> Order(S2);
3
gap> espacio(S2,u); #Plane.
2

gap> S2:=(R3*R1*R2)^2;;
gap> Order(S2);
5
gap> espacio(S2,u); #Plane.
2
```

Bibliography

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